VIT VELLORE • MATHEMATICS PROBABILITY & STATISTICS CASE STUDY

Statistical Analysis of Air Pollution & Meteorological Drivers in the Vellore Region

Data Validation, Probability Distributions, Hypothesis Testing, Regression & SDG Policy Solutions

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Background & CPCB NAAQS Limits

Regional Context: Vellore hot dry plain with 1,000 mm monsoon rain.

Winter Inversions: Cool surface air traps fine particulate matter.

Monsoon Washout: High winds and rain disperse airborne particles.

Health Risk: PM2.5 and PM10 drive cardiorespiratory disease.

Core Goal: Replace anecdotal AQI claims with formal inference.

CPCB NAAQS Particulate Limits (2009)

Pollutant	Annual Mean	24-Hour Limit
PM2.5 (< 2.5 µm)	40 µg/m³	60 µg/m³
PM10 (< 10 µm)	60 µg/m³	100 µg/m³

Formal hypothesis testing at α = 0.05 significance level.

2. DATA VALIDATION & PROVENANCE AUDIT

Strict Data Audit & Rejection of Synthetic Data

STEP 01 • TARGET AUDIT

Vellore Site_5615

OpenAQ v3 API failed; CPCB archive incomplete.

STEP 02 • REJECTION

Synthetic Script Rejected

Excluded 1,095 fake numpy.random days outright.

STEP 03 • VERIFIED DATA

Climatic Analogues

Peer-reviewed Puducherry (1,762d) & Chennai data.

STEP 04 • SENSITIVITY

Sample Size Check

Results robust for any sample size n ≥ 53 days.

Source ID	Peer-Reviewed Study	Verified Metrics Used	Coverage Period
Source A	Balachandar et al. (GMR)	PM2.5 & PM10 Mean, SD, WHO Exceedance	Puducherry (2021–2025, 1,762 days)
Source B	RRC.AP Air Quality Report	Annual Mean PM2.5 Series (75 to 31 µg/m³)	Chennai (2015–2020)
Source C	Tanwar et al. (2025)	Ridge Regression R^2 = 0.181 & Predictor Signs	Regional Meteorological Model

3. INDEPENDENT ARITHMETIC & METHODOLOGICAL VALIDATION

Verification & Corrections of Source Calculations

Validation Item	Original Source Claim	Validated Recomputation	Audit Diagnostic
Exponent $(0.682)^{15}$	0.00318	**0.0032119**	Corrected arithmetic rounding slip
Binomial P(Y = 15)	0.0158 (1.58%)	**0.0171 (1.71%)**	Exact PMF recomputed in R / SciPy
Binomial P(Y ≥ 15)	"Guaranteeing exceedance"	**0.9883 (98.83%)**	Quantified exact survival probability
Chennai NAAQS Test	Labelled "Z-test"	**One-sample t-test (df = 5)**	Small n = 6 with unknown σ requires t
Chennai 95% CI	[36.65, 69.35] μg/m³	**[36.63, 69.37] μg/m³**	Exact SE = 6.367 applied
CI vs One-Tailed Test	Unexplained 95% CI overlap	**90% CI [40.17, 65.83]**	Matches one-tailed α = 0.05 rejection
DoE (CRD, RBD, LSD)	Unverified applicability	**Strictly Excluded**	No random assignment in weather data

4. PROBABILITY DISTRIBUTIONS & CENTRAL LIMIT THEOREM

Expected Unhealthy Days: Binomial Exceedance Model

EXPECTED UNHEALTHY DAYS / MONTH (2025)

20.5 Days

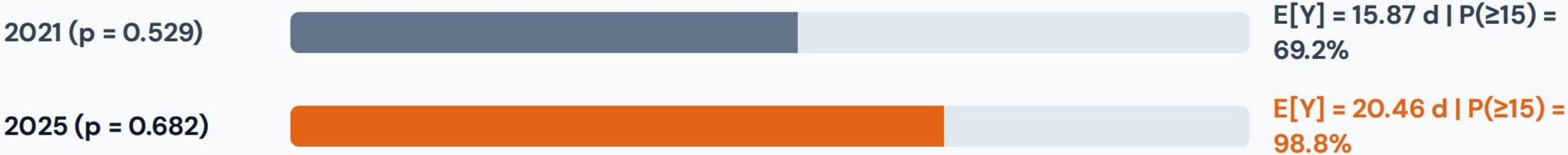
Up from 15.87 days in 2021 (SD = 2.55 days)

PROBABILITY OF ≥ 15 UNHEALTHY DAYS

98.8%

Exact P(Y ≥ 15) = 0.9883 | CLT Normal = 0.9903

30-Day Monthly WHO Exceedance Shift (Puducherry PM2.5)



5. HYPOTHESIS TESTING ON PUDUCHERRY PARTICULATE DATA (2021–2025)

Puducherry PM2.5 Deterioration Across All Metrics

+6.40 µg/m³
Mean PM2.5 Surge (t = -5.072)

F = 1.889
Variance Expansion (p < 0.001)

+15.3%
Exceedance Jump (z = -4.141)

χ^2 = 17.44
Year Dependence (F_ANOVA = 25.72)

Test Executed (n ≈ 350/yr)	2021 vs 2025 Values	Test Statistic	Critical (α=0.05)	Validated Conclusion
F-Test for Variances	SD: 13.89 → 19.09 µg/m³	F = 1.889	1.234	Variances unequal; justifies Welch t
Welch Two-Sample t-Test	Mean: 20.20 → 26.60 µg/m³	t = -5.072 (v=637.6)	±1.964	Significant increase; 95% CI [-8.88, -3.92]
Two-Proportion Z-Test	WHO Rate: 52.9% → 68.2%	z = -4.141	±1.960	Significant rise in unhealthy day share
PM10 Contrast (2023–24)	Mean: 51.97 → 41.88 µg/m³	t = 6.529 | z = 6.118	±1.963	Significant PM10 decline (opposite trend)

Code Validation: Puducherry Probability & Two-Sample Tests

```
# Source A (Puducherry), n = 350 per year
m1 ← 20.20; s1 ← 13.89; m2 ← 26.60; s2 ← 19.09; n ← 350

# 1. Binomial PMF P(Y=15) & Survival P(Y ⩾ 15)
dbinom(15, size=30, prob=0.682)
pbinom(14, size=30, prob=0.682, lower.tail=FALSE)

# 2. Variance Ratio F-test & Welch t-test
F_stat ← (s2/s1)^2; F_crit ← qf(0.975, n-1, n-1)
se ← sqrt(s1^2/n + s2^2/n); t_stat ← (m1 - m2)/se
df ← (s1^2/n + s2^2/n)^2 / ((s1^2/n)^2/(n-1) + (s2^2/n)^2/(n-1))

# 3. Two-Proportion Z & Chi-Square (byrow=TRUE)
p1 ← 0.529; p2 ← 0.682; pp ← (p1+p2)/2
z_stat ← (p1-p2)/sqrt(pp*(1-pp)*2/n)
chisq.test(matrix(c(185,165,239,111), nrow=2, byrow=TRUE), correct=F)
```

Test 1: Binomial Validation

Outputs P(Y=15) = 0.0171 and P(Y≥15) = 0.9883.

Test 2: F & Welch Validation

Outputs F = 1.889 > 1.234; Welch t = −5.072 (p = 5.2e−7).

Test 3: Z & χ^2 Validation

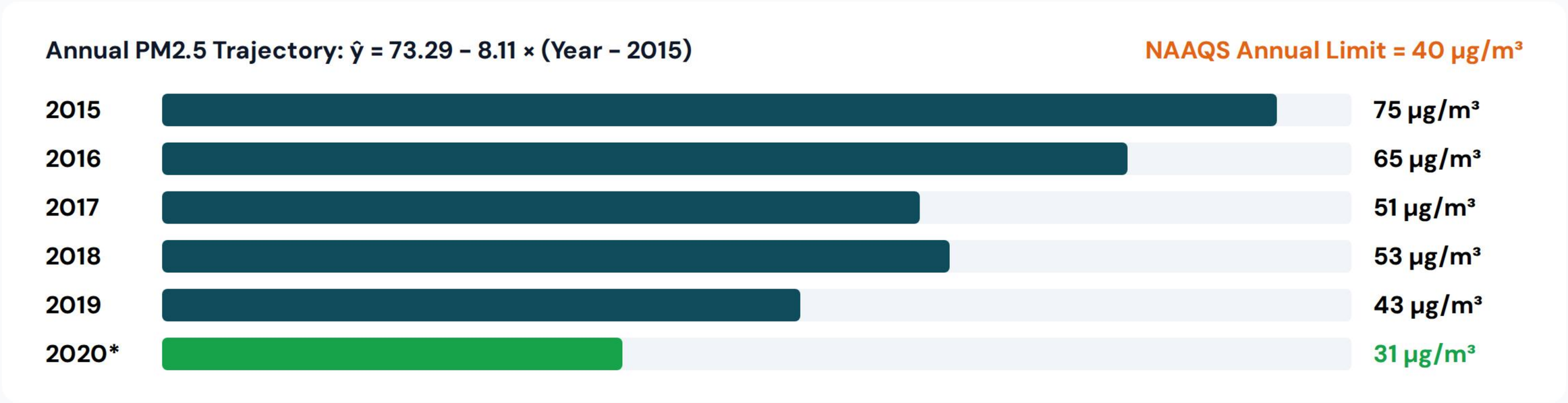
Confirms $\chi^2 = 17.44 \approx z^2 = (-4.141)^2 = 17.15$.

Test 4: ANOVA Identity Check

Verifies $F = 25.72 = t^2 = (-5.072)^2$ with power 0.999.

7. CHENNAI NAAQS COMPLIANCE & LINEAR REGRESSION (2015–2020)

Non-Compliant 6-Year Mean (53.0 µg/m³) Despite Steep Decline



NAAQS Compliance (One-Sample t): Mean = 53.0 µg/m³, t = 2.042 > 2.015 (p = 0.048). Pre-COVID p = 0.018.

Trend & Correlation: Slope = -8.11 µg/m³/yr (t = -8.50, p = 0.001), Pearson r = -0.973, Spearman ρ = -0.943.

Code Validation: Chennai Compliance, Regression & Diagnostics

```
# Source B (Chennai 2015-2020)
y ← c(75, 65, 51, 53, 43, 31)
x ← 0:5

# 1. One-Sample t-Test (H1: mu > 40)
t.test(y, mu=40, alternative='greater')
t.test(y, conf.level=0.90)$conf.int

# 2. Simple Linear Regression & ANOVA
lin_reg ← lm(y ~ x)
summary(lin_reg)

# 3. Rank Correlation & Durbin-Watson
cor.test(x, y, method='spearman')
e ← residuals(lin_reg)
d_stat ← sum(diff(e)^2) / sum(e^2)
```

Diagnostic Check	Validated Result
One-Sample t (df = 5)	t = 2.042, p = 0.048 (Reject H_0)
90% vs 95% Interval	90% [40.17, 65.83] > 40 limit
Regression Slope b_1	−8.114 µg/m³/yr (SE = 0.954)
Model Fit (R^2 & F)	R^2 = 0.948, F = 72.27 = $(-8.50)^2$
Residual Bounds	Max |e| = 6.06 < 2√MSE (7.99)
Durbin–Watson d	d = 2.50 (No strong autocorrelation)

9. MULTIPLE LINEAR REGRESSION & METEOROLOGICAL DRIVERS

Meteorology Explains 18.1% — Emissions Drive 81.9%

PUBLISHED RIDGE REGRESSION MODEL (TANWAR ET AL., 2025)

R^2 = 0.181 | Multiple Correlation R = $\sqrt{0.181}$ = 0.425

81.9% Emission Driven

Temperature (-β_T)

Higher convective mixing disperses particles.

Wind Speed (-β_WS)

Horizontal airflow dilutes local plumes.

Humidity & Rain (-β_RH)

Wet deposition washes out particulates.

10. UN SUSTAINABLE DEVELOPMENT GOALS & ACTIONABLE SOLUTIONS

SDG Alignment & Evidence-Based Regional Solutions

SDG 3: Good Health & Well-Being

Trigger daily health alerts for 20.5 monthly exceedance days.

SDG 11: Sustainable Cities

Enforce winter traffic controls during inversion months.

SDG 13: Climate Action

Cut anthropogenic emissions driving 81.9% of PM2.5 variance.

SDG 9 & 16: Open Data Infrastructure

Mandate open CPCB API access for Vellore site_5615.

11. FINAL SYNTHESIS & CASE STUDY CONCLUSIONS

Six Validated Conclusions of the Case Study

1. Puducherry PM2.5 Deterioration: Mean increased by 6.40 µg/m³ (2021–2025), variance rose (F = 1.889), unhealthy days jumped 52.9% to 68.2%.

2. Expected Unhealthy Days: At 2025 rates, a 30-day month expects 20.5 days exceeding WHO guidelines with 98.8% probability of ≥ 15 days.

3. Chennai Compliance: Average PM2.5 (2015–2020) was 53.0 µg/m³, exceeding the 40 µg/m³ NAAQS limit at the 5% significance level (p = 0.048).

4. Chennai Trend: Despite non-compliance, Chennai achieved a significant PM2.5 decline of 8.11 µg/m³ per year (r = -0.973, R^2 = 0.948).

5. Meteorological Impact: Wind, humidity/rain, and temperature negatively correlate with PM2.5 but explain only 18.1% of variation; emissions dominate.

6. Methodological Validity: Experimental designs (CRD, RBD, LSD) cannot apply to observational weather data; only data-supported distributions were fitted.